

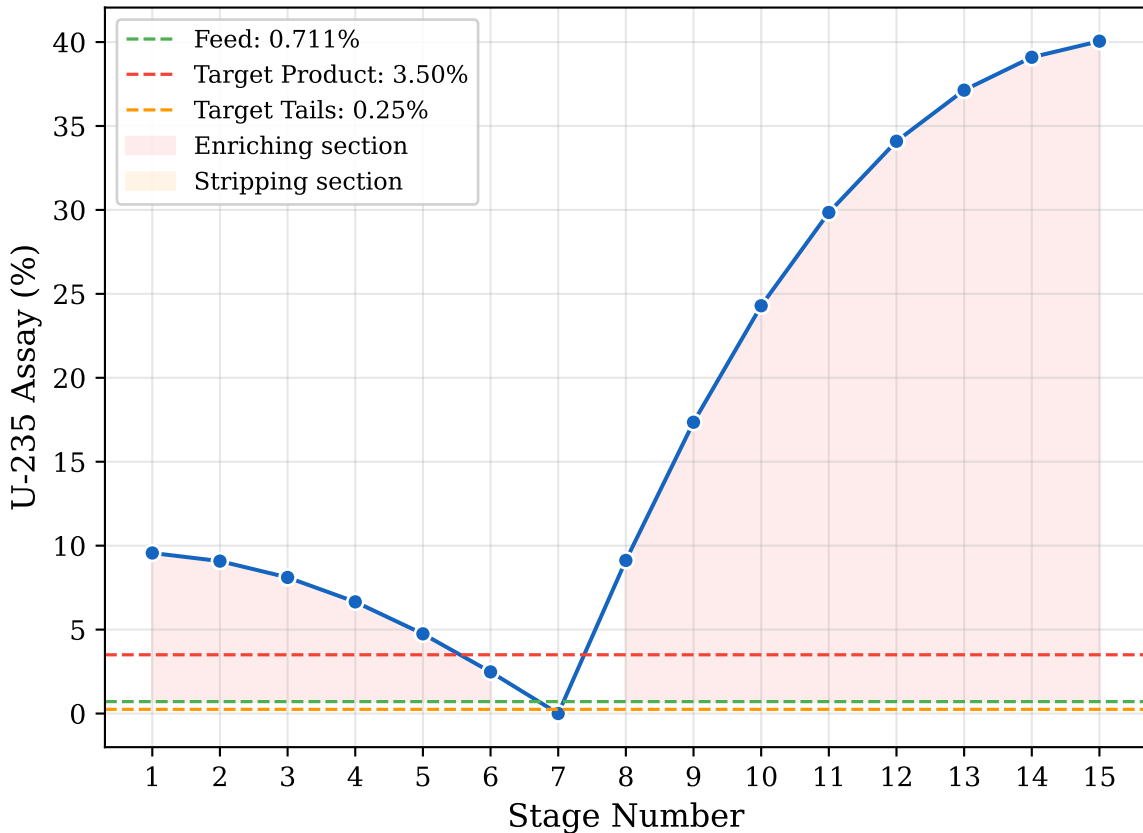
Gas Centrifuge Cascade Simulation

Enrichment Analysis Report

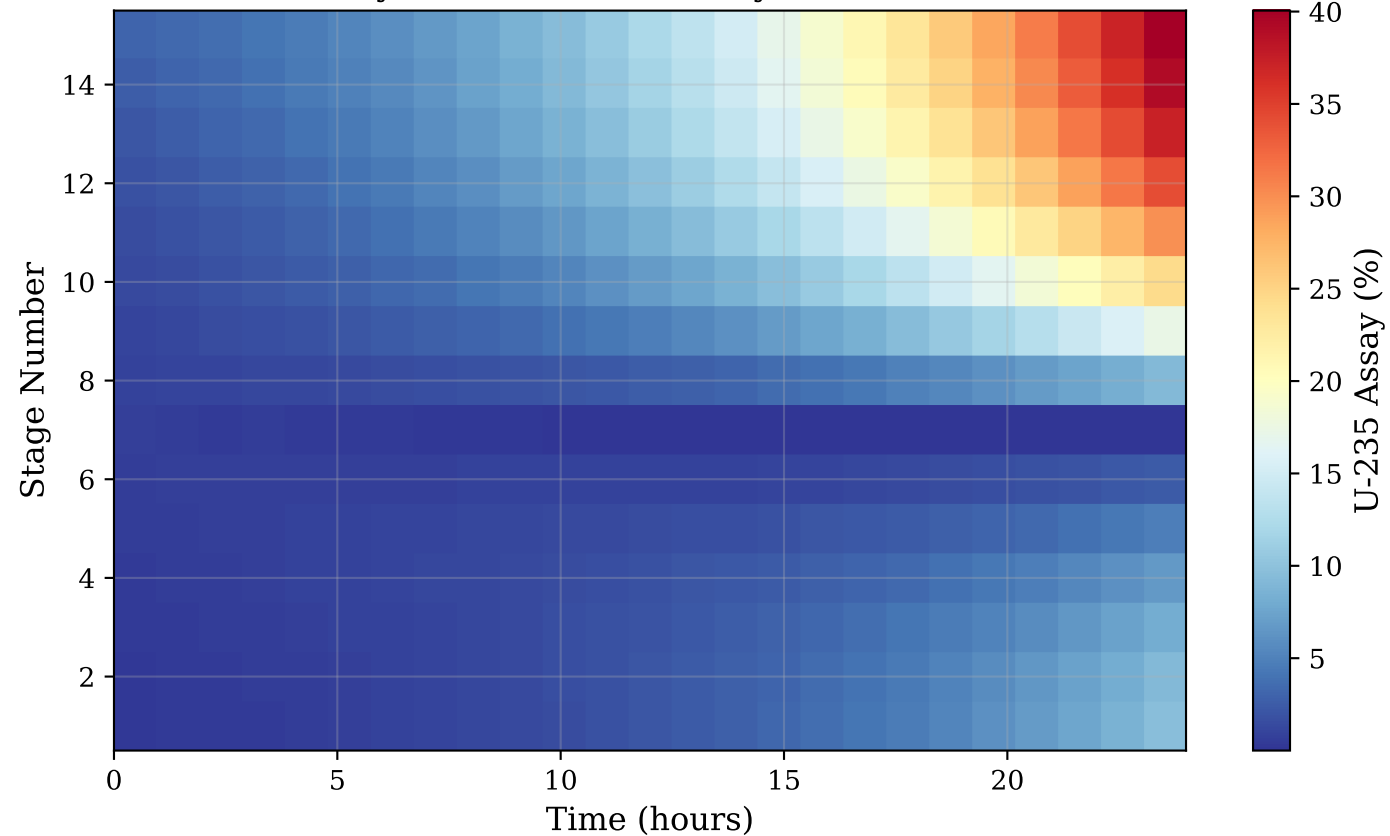
March 12, 2026

Generated by Nuclear Fuel Cycle Simulation Framework

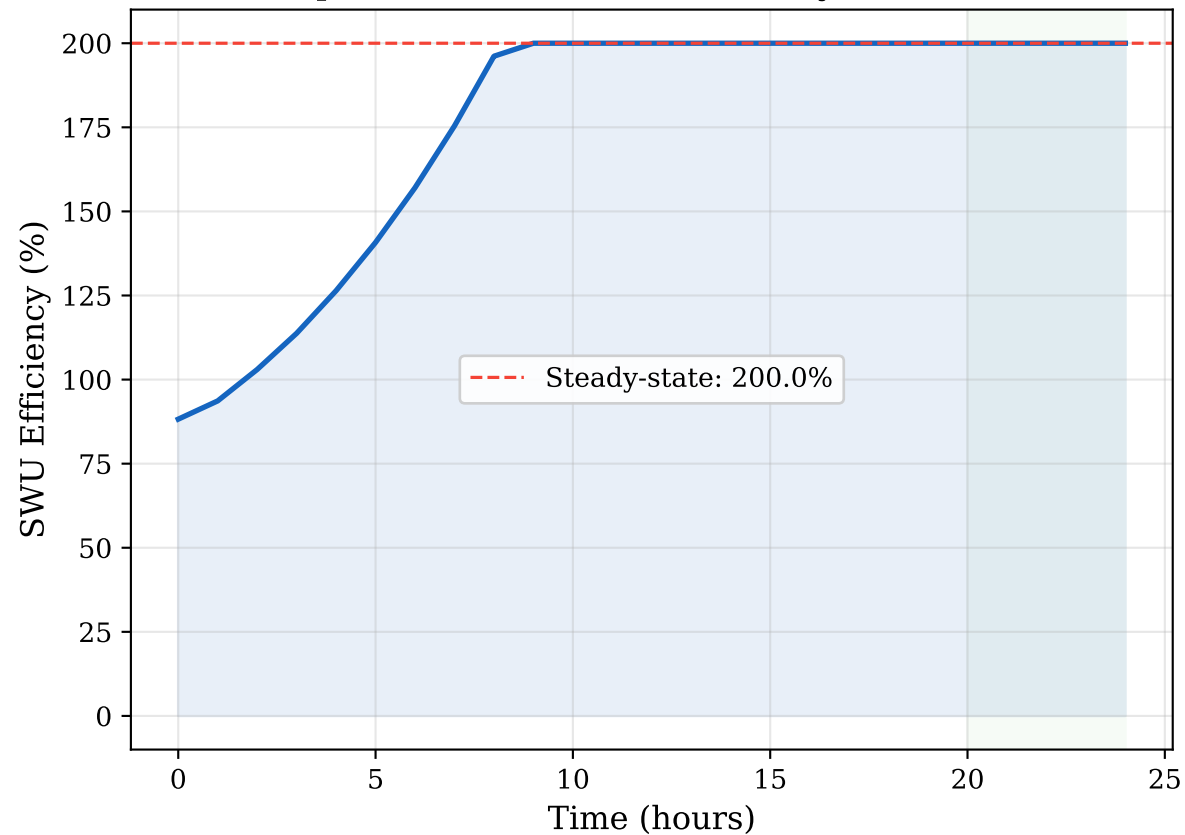
Cascade Stage Assay Profile (Steady State)



Dynamic Cascade Assay Evolution



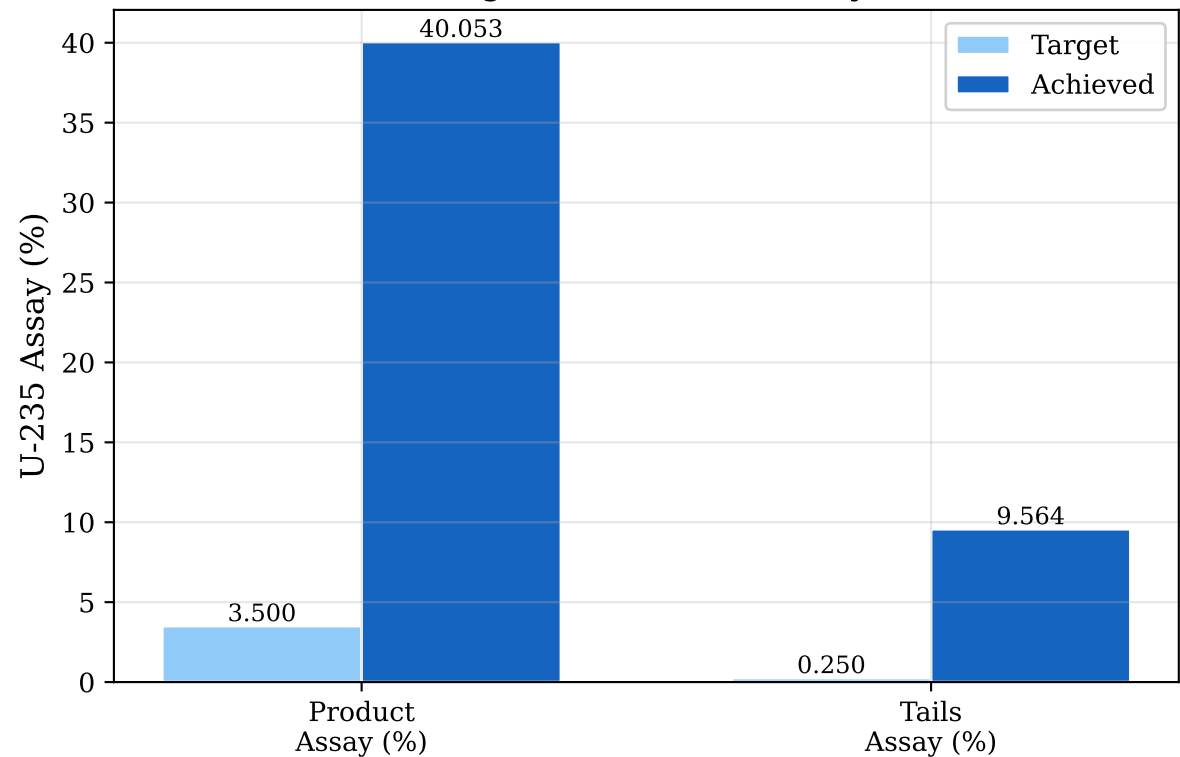
Separative Work Unit Efficiency Over Time



Gas Centrifuge Cascade Performance

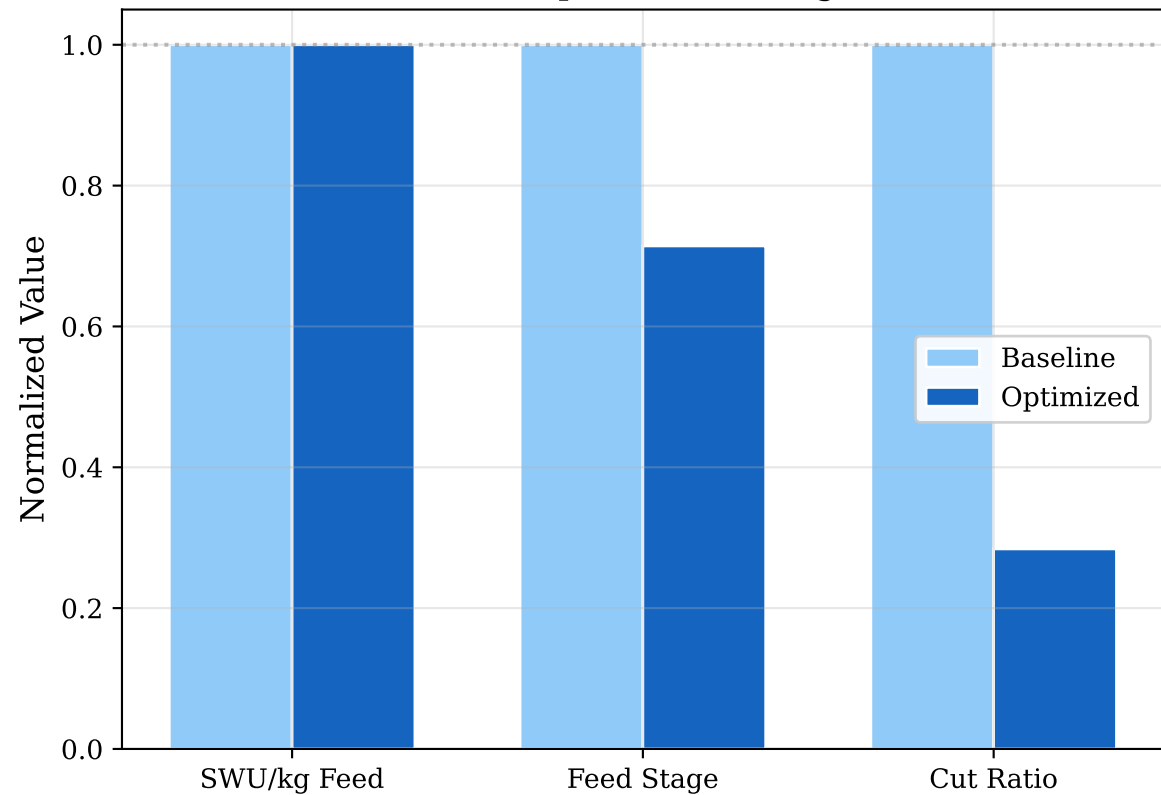
Cascade Performance Metrics

Target vs Achieved Assay

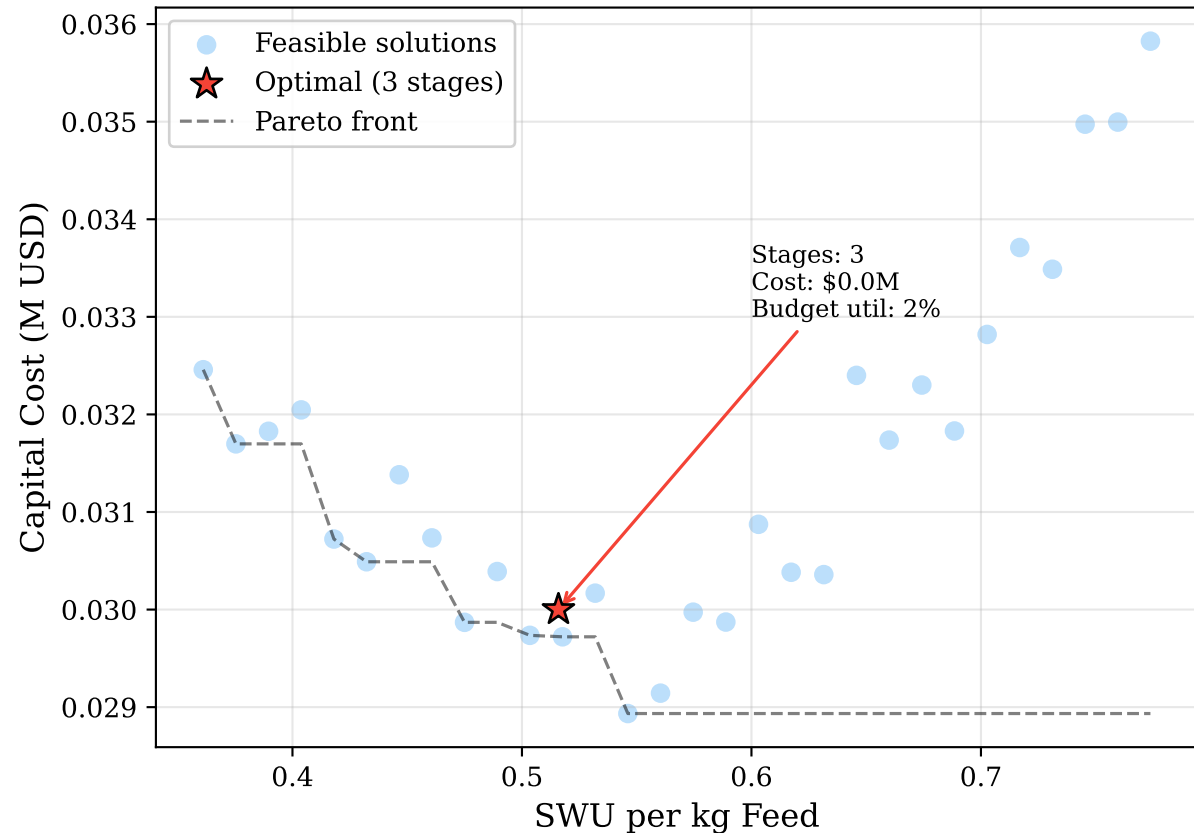


Metric	Value
Product Assay (achieved)	40.0525%
Tails Assay (achieved)	9.5642%
Separation Efficiency	200.00%
Total SWU	1637.78
SWU per kg Feed	0.6824
Number of Stages	15
Machines	1,500

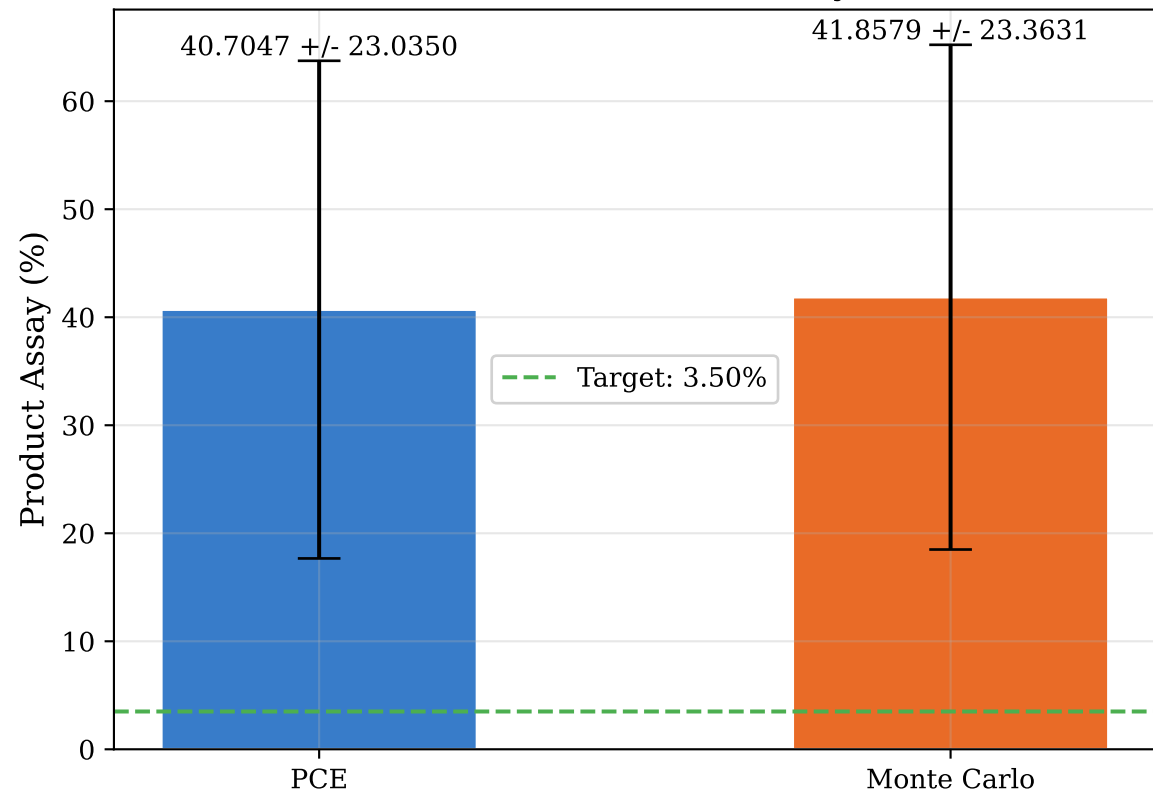
Baseline vs Optimized Configuration



Multi-Objective Optimization: SWU vs Cost



PCE vs Monte Carlo: Product Assay Prediction



Product Assay Uncertainty Distribution (PCE)

